

Does Avatar Photorealism Enhance Social Presence? A Proposed Study Comparing Captured, Crafted, and Stylized Models

Maryam Kameli*

Praneet Bala†

Victoria Interrante‡

University of Minnesota



Figure 1: 3D Gaussian Splatting now makes photorealistic capture of a specific person possible. Here a real person (top) is reconstructed as a captured 3DGS avatar (bottom) that keeps the subject’s identity, expression, and pose across held-out views. We plan to use VR to compare users’ experience of the captured avatar vs a crafted MetaHuman of the same person and assess the extent to which image metrics can predict the differences.

ABSTRACT

Over the past decade, the process of creating realistic animatable avatars has progressed from reliance on textured parametric meshes to the use of neural radiance fields. Most recently, 3D Gaussian Splatting (3DGS) has made it possible to build photorealistic avatars from a dense series of images. Now that achieving true photorealism is finally possible, we need to re-address the question of whether, and when, deploying truly realistic avatars improves the human experience. Some VR literature suggests it may not: behavioral realism has consistently been shown to have a stronger impact on social outcomes than visual realism. Stylized avatars have often performed as well as realistic ones, and the relation between visual realism and user experience is widely understood to be non-monotonic, with the highest levels of realism triggering worse outcomes due to the uncanny valley. However, these conclusions were drawn from research predating the ability to achieve avatars whose visual quality is indistinguishable from a photograph. We propose a second-person VR experiment to answer this question. We will create three avatars of the same confederate: one captured and rendered via Gaussian Splatting, one crafted using Epic Games’ MetaHuman pipeline, and one stylized avatar. Then, we will measure co-presence, interpersonal trust, eeriness, and perceived visual realism as participants view each avatar. We will also test whether image fidelity metrics (PSNR, LPIPS, SSIM) can predict any of the experiential outcomes. In this paper, we describe a pipeline that

produces the avatars and report its current fidelity.

Index Terms: Virtual Reality, 3D Gaussian splatting, Photorealistic avatars, Neural rendering, Social presence, Avatar realism, Perceptual evaluation, Telepresence, Sense of embodiment, Radiance fields, MetaHuman, Uncanny valley, Avatar-mediated communication

1 INTRODUCTION

Building a 3D, animatable avatar of a person from images or a video is one of the oldest goals in computer vision and graphics. The 3D Morphable Model [4] is the precursor to several recent lines of work: FLAME [28], SMPL [31], avatar creation from hand-held video [22] and from single images [21]; the neural-field era of NeRF [36], monocular facial NeRFs [16], and the deep appearance models behind Codec Avatars [30]; and most recently 3D Gaussian Splatting (3DGS) [25]. GaussianAvatars [40] binds 3D Gaussian to a triangle of a FLAME mesh, so that changing the expression and pose parameters deforms the splats with the mesh. This yields a photorealistic, drivable head avatar trained from multi-view video. Appearance fidelity matters in immersive settings because the avatar is the interface: it carries the user’s identity, mediates social contact, and shapes presence and embodiment. Yet, in the computer vision literature, progress is reported almost exclusively in terms of image-fidelity metrics such as PSNR, SSIM, and LPIPS. These metrics measure how closely a rendered image reproduces a ground-truth image; they say nothing about the extent to which a user’s engagement with the avatar matches their experience of engaging with a real person, how co-present they feel with it, to what extent they trust it as a social partner. The XR community faces two significant questions: which aspects of realism actually matter for human experience, and when is it worth bringing photorealistic assets into MR/AR? Some VR research has concluded that visual realism is not a dominant factor in how a virtual environment is

*e-mail: kameli026@umn.edu

†e-mail: balax007@umn.edu

‡e-mail: interrante@umn.edu

experienced [5, 34, 27]. Those studies, however, predate today’s state-of-the-art photorealistic modeling and rendering capabilities, and the field still lacks any validated link between the image-fidelity metrics graphics reports and the experiential measures VR collects. 3DGS now provides the missing tool. We address this question with a study design that compares multiple approaches to creating an avatar of the same real person and measures the human outcomes, with the real person as a control:

- a captured avatar: a radiance-field (3DGS) reconstruction that re-creates the person’s appearance from their capture.
- a crafted avatar: a state-of-the-art parametric model (Epic MetaHuman) [10] that is built to resemble the same person, and rendered with the greatest level of realism achievable in current VR game engines.
- a stylized avatar: a cartoon-style avatar (Meta/Horizon) built from a phone capture, which makes no attempt at photographic likeness and represents what a very large share of social VR users see today.

We will ask participants to separately consider each avatar representation, and to report on their sense of co-presence and interpersonal trust, as well as on the model’s apparent visual realism, identity fidelity, and eeriness. We will then relate these outcomes to the image-fidelity metrics.

Contributions

- We argue that findings about the impacts of avatar realism that were obtained using semi-realistic models cannot not be extrapolated to situations in which the avatar is perceptually indistinguishable from a real person.
- We describe the steps that can be taken to adapt a studio-tuned photoreal capture pipeline to unconstrained, in-the-wild room capture. These include: camera re-aiming to satisfy the centered-principal-point assumption of the 3DGS rasterizer, and the use of a single-resampling image chain to restore detail. Finally, we characterize the capture-bounded fidelity afforded by this approach.
- We outline the design of a user study comparing co-presence, inter-personal trust, perceived visual realism, identity fidelity, and eeriness between a captured 3DGS avatar and a crafted MetaHuman and stylized Meta avatar of the same person. Additionally, we evaluate the correlation between traditional image-fidelity metrics and these user experience metrics.

2 BACKGROUND AND RELATED WORK

Avatars can be generated using parametric meshes [4, 31, 28], which are relatively straightforward to animate. However, these models are prone to having have a texture-mapped appearance, and achieving photorealistic-looking hair, eyes and mouth interiors is challenging. Surface details captured using video [22] or even a single photograph [21], can be used to improve the model’s appearance, but the underlying mesh representation limits what can be achieved. Ichim et al. [22] omit hair, while Hu et al. [21] reconstruct hair from one image, but neither overcome the blend-shape template’s limitations in accurately representing mouth interiors and complex, view-dependent expressions. NeRF [36] and NeRFace [16] learn a continuous radiance field mapping spatial coordinates and viewing directions to color and volume density. This implicit representation can better preserve fine details, but is more costly to render. The original NeRF took around thirty seconds to produce a single frame on a modern GPU [36], but a VR headset needs to refresh at 90 Hz and maintain a motion-to-photon

latency of under about 20 ms to ensure a comfortable user experience [33, 41]. Lombardi et al.’s deep appearance models [30], which are the foundation of Codec Avatars [32], achieve photorealistic and view-dependent faces that render in real time, but they rely on dense, synchronized data collected inside a dedicated multi-camera capture dome. While earlier methods had to choose between speed and quality; 3D Gaussian Splatting [25] does not force this compromise. In 3DGS, a scene is stored as a set of 3D Gaussians; each has a position, an anisotropic covariance, an opacity, and a view-dependent color encoded with spherical harmonics. The Gaussians are rasterized and alpha-composited on the GPU, so rendering is fast enough for real-time while maintaining quality comparable to the best neural radiance fields before it. GaussianAvatars [40] made the static representation animatable by binding each Gaussian to a triangle of a FLAME mesh, so the splats deform as the mesh is posed. And later, other systems have brought 3DGS avatars into game engines and onto mobile and AR hardware [50, 7]. The weak point with this approach is extrapolation; if the head turns to an angle that was unseen during training, misaligned Gaussians manifest as floating artifacts that pop into and out of view. In contrast, extrapolated mesh representations degrade more gracefully, simply appearing stretched or blurred at extreme angles. It remains to be determined whether viewers prefer the higher peak fidelity of 3DGS avatars despite their potential for severe artifacts, or the safer but less realistic mesh-based alternative.

Presence in a virtual environment is typically understood as a combination of place illusion and plausibility [44] and can be assessed using instruments like the Igroup Presence Questionnaire (IPQ) [42]. The sense of embodiment in an avatar is traditionally conceptualized in terms of body ownership, agency, and self-location, and assessed using different standardized instruments [26, 18]. The appearance of a user’s own self-avatar can affect their behavior in VR (the Proteus effect [49]), and disembodied users have been shown to face higher cognitive demands during spatial tasks [46]. A comparison between photogrammetric and wooden self-avatars found a stronger sense of human likeness and body ownership in the higher realism condition, but also a higher sense of eeriness and no benefit in spatial presence [27].

Appearance fidelity is another aspect of visual realism. Interviews with users of social VR find that many people want to be embodied in an avatar that resembles their real self [14]. In social VR, the appearance of an interlocutor’s avatar can also affect co-presence and social presence: the feeling of being together with another person and sharing a state of mutual understanding [3, 2]. Aseeri et al. [1] found that users reported a greater sense of inter-personal trust, but equal levels of co-presence, when engaging with a confederate who was represented with perfect fidelity in video-see-thru mode versus as a photogrammetrically scanned and animated avatar.

The realism of an avatar depends not only on what it looks like but also on how it behaves. Prior work suggests that high visual realism demands high behavioral realism, and that failing to match a realistic appearance with equally realistic behavior can degrade outcomes relative to when the appearance fidelity is lower [17]. A substantial line of work holds that behavioral realism, meaning plausible gaze, motion, and responsiveness, matters more than photographic realism for social response [5, 17]. However, strong behavioral realism does not ensure strong social presence when visual realism is very low [2].

The current research consensus seems to argue against the assumption that increasing appearance realism generally yields increasing benefits. Theoretical work argues that plausibility, not photorealism, is the pivotal condition for realistic response [45], and co-presence has been shown to respond more to motion synchronization than to the completeness of the virtual body [19]. Studies have shown that fully stylized characters can be as ap-

peeling and as trustworthy as more realistic ones, and highlight an eeriness penalty when visual realism attempts to be high but misses [34, 47]. A recent immersive study also reports higher co-presence with a stylized avatar than with a realistic one [9]. Further supporting this notion, Fraser et al. [13] had 71 participants disclose personal experiences in VR to a motion-captured interviewer appearing as a cartoon avatar, a generic social-VR avatar, or a MetaHuman, and found no significant effect of avatar type on perceived realism or enjoyment immediately after the interactions, although the MetaHuman was most often ranked the most realistic at the end.

On the other hand, Kaiser et al. [24] report the opposite pattern: in augmented reality, across avatars ranging from abstract representations to a realistic point-cloud reconstruction, higher realism generally enhanced social presence and comfort. Likewise, Seymour et al. [43] found that users reported higher levels of affinity and trustworthiness when interactively engaging, in both 2D and VR, with a photorealistic digital human versus a stylized cartoon caricature. This led them to argue that the uncanny valley can now be crossed.

What is most relevant to our study in all of this is that the vast majority of the research that informs our present understanding of the benefits of avatar realism (or the lack thereof) predates the ability to present users with 3D avatars that are visually indistinguishable from real people. Findings of little gain or loss in the comparison between an avatar of modest realism vs an avatar of low realism, or between a stylized avatar and an imperfect scan, cannot be extrapolated to the conditions of true photorealism that we are on the cusp of being able to now achieve. This last point highlights the importance of using more precise metrics to quantify the realism of an avatar’s appearance.

In computer vision, quality is reported in terms of PSNR/SSIM/LPIPS against held-out views. A recent comparative benchmark of smartphone-captured neural avatars with conventional mesh avatars found that the latter outperformed the former with respect to artifacts, novel-view and novel-pose fidelity, and rendering performance. The authors report that while the neural avatars performed well near their training views, away from this their quality quickly degrades to a level that is “not acceptable for social VR applications” [35]. Nevertheless, we anticipate that current best approaches may suffice in specific restricted contexts.

2.1 The two routes to a personal avatar

We plan to compare personalized avatars obtained from three production routes. The captured one (3DGS/GaussianAvatars) reproduces the person’s appearance from their own imagery; its fidelity is bounded by the capture, so it inherits the capture’s artifacts as well. The crafted one is built with Epic MetaHuman [10], which provides a parametric, riggable head that can be made to resemble a person (for example, via Mesh-to-MetaHuman from a scan); this route produces clean geometry and animation, but the appearance will not be identical to that of the real person. The third is a stylized avatar (Meta/Horizon) built from a phone capture, which makes no attempt at photographic likeness and marks the stylized end of the range. All can be deployed as VR-ready assets. Which one gives a user a better experience is an open question, and this is the question our research aims to address.

3 THE CAPTURED AVATAR

Our capture pipeline is still under development. The reconstruction method we use is unchanged from GaussianAvatars [40], and we claim no algorithmic novelty. Our contribution is in reporting on what it takes, and what it costs, to run that method on an ordinary room capture rather than the studio rig it was designed for. We present it here as a proof of concept and describe its current fidelity, as it is central to our argument. **Capture** A synchronized rig of

25 ~4K cameras records the subject performing a sequence of actions spanning range-of-motion and speech: a neutral pose, jaw and mouth extremes, brow and eye actions, the canonical emotions, and connected speech. Together, these cover the expression space from many viewpoints. We use an *in-the-wild room capture*, though, with the cameras framing the upper body, rather than a face-filling studio close-up.

Reconstruction Every method we use at this stage is based on prior work. We synchronize the frames and calibrate the cameras to a shared pinhole model. We then segment the subject with Robust Video Matting [29] and recover per-frame FLAME [28] parameters (identity, expression, pose) with the photometric tracker released alongside GaussianAvatars [40]. Finally, we optimize a GaussianAvatars model that binds 3D Gaussians to FLAME triangles, so that the avatar is drivable with the same parameters used to fit it. For the details of these methods, we refer the reader to the original papers.

The given pipelines were predicated on a studio rig, in which every camera points directly at the face, and the face fills most of the frame for each camera. Our capture setup satisfies neither assumption, with the result that the head occupies only a small patch within each image.

The 3DGS rasterizer assumes a centered principal point, meaning the camera looks straight down the middle of the frame. A plain crop around our off-center head breaks that assumption. It moves the principal point off-center, so the head falls outside the symmetric frustum that the rasterizer draws, and nothing renders. Our fix is to aim each camera at the FLAME head centroid using a homography derived solely from camera rotation. This gives a training set in which the head is centered and fills the frame, and it leaves the FLAME fit in world space untouched.

The reconstruction expects views centered on the face. Getting there from a capture that frames the whole body has been the main practical obstacle to bringing such captures into an engine. Two details of the warp mattered more than we expected. First, the virtual cameras have to aim at the tracked mesh’s vertex centroid, not its root joint. Second, the images should be resampled once. We composed the lens distortion correction and re-aimed warp into a single sampling step and applied it directly to the raw frames. Compared with our first version, which resampled in two passes, this raised the sharpness of the training images by 44% (Laplacian variance over foreground pixels) and restored the eyes and eyebrows in the reconstruction.

Deployment We export the trained model as a standard 3DGS point cloud and render it in Unity. The avatar is drivable through its FLAME parameters, which we currently demonstrate in the desktop viewer released with GaussianAvatars [40]. We will bring the FLAME parameter control into Unity [48] for the VR study as the next step.

Current fidelity Because the subject occupied only a small part of the frame, the head carries on the order of 200 native pixels. That is roughly an order of magnitude below the face resolution of a studio face rig. The avatar is recognizable and animates correctly under FLAME control, but it is visibly soft. And in views unseen during training, it shows the classic popping and floaters. On unseen data, the trained model reaches PSNR 22.1, SSIM 0.87, and LPIPS 0.10. The nearest published reference point is GaussianAvatars itself, which reports PSNR 26.0, SSIM 0.910, and LPIPS 0.076 under the matching protocol, driving the avatar with a held-out sequence of the same subject, on NeRSemble studio captures where sixteen cameras frame the head and shoulders at 802×550 [40]. Our perceptual gap is modest (LPIPS 0.10 against 0.076), and the larger pixel-metric gap tracks the difference in native head resolution, which is consistent with the capture, rather than the method, being the binding constraint. Quality also depends heavily on viewpoint: measured on the face region alone, well-covered viewpoints reach about 25.5 dB, while oblique views that the training cameras



Figure 2: Comparison of the real person (top) and the captured 3DGS avatar (bottom) across seven poses and viewpoints. Under each column, we give image PSNR, SSIM, and LPIPS, plus a PSNR for the face region alone.

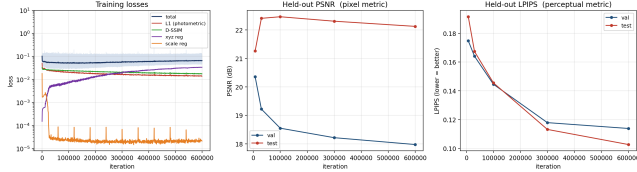


Figure 3: Training behavior of the captured avatar over 600k iterations. Left: the total loss and its terms on a log scale. Middle: held-out PSNR. Right: held-out LPIPS.

barely covered fall to 17.6 dB (Fig. 2).

Building the system also showed us the metric disconnect in miniature, twice. During training, PSNR on the unseen frames peaks early and then declines, while LPIPS continues to improve (Fig. 3), so a pixel metric and a perceptual metric rank the same checkpoints in opposite directions. A second instance came from the switch to resampling only once. It visibly resolved the irises and eyebrows that the earlier model had smeared; however, the aggregate numbers barely moved (PSNR 22.1 against 22.2, LPIPS 0.10 in both cases). Part of the reason is that the sharper ground truth is also a harder target. Doubling the training schedule slightly worsened perceptual quality. Rendering the training set again at a tighter zoom, which magnifies interpolation blur and calibration error, also degraded every measure. The binding constraint is the capture itself, so we are capturing again with a framing in which the head fills most of the frame. This limitation yields a captured avatar that is authentic but soft, set against a crafted avatar that is sharp but synthetic. It also makes concrete that the image metrics alone cannot tell us which one a viewer will accept as a present partner rather than an eerie one.

4 THE CRAFTED AVATAR

We built the crafted avatar as an Epic MetaHuman [10] from a scan of the same person (Fig. 4). We first captured the person with a photogrammetry app (Polycam) [39]. We cleaned the mesh in Blender [6] and repaired the texture. We then exported the mesh to Unreal Engine [11], created a MetaHuman Identity from the scan, and ran the Identity Solve, which fits Epic’s template head to the scanned geometry using tracked facial landmarks. The Mesh-to-MetaHuman step returned a rigged head that keeps the scanned face shape and carries MetaHuman’s own synthesized skin texture. We set the skin tone by hand. The result is an editable head that matches the real face shape while its surface appearance comes from MetaHuman rather than from the scan. The available hair assets were limited, and none was a good substitute for the model’s hair; we chose the closest.

The stylized avatar We also built a Meta avatar of the same person to use as a third comparator (Fig. 5). We created it using Meta’s avatar tool from a phone capture of the same model’s face, which the tool turns into a stylized avatar for Horizon Worlds with no



Figure 4: Steps in building the MetaHuman avatar. The top row shows the raw Polycam scan imported into Blender (left) and the cleaned face mesh prepared for Unreal Engine (right). The middle row shows the segmented reference photograph beside the template head fitted to it during the identity solve. The bottom row shows the identity solve with tracked facial landmarks on the scan (left) and the resulting MetaHuman in Unreal Engine (right).

attempt at photographic accuracy. This marks the stylized end of the fidelity range we plan to compare and represents what a very large share of social VR users see today.

5 PROPOSED EXPERIMENT

We propose a within-subjects study that isolates *visual realism* by presenting the same person as three static avatars of differing realism levels, each shown in a matched environment, and asking participants to judge them. We plan to start with *static* avatars, so there is no avatar behavior to control in the present design; in follow-on work we plan to implement dynamic avatars driven by live puppeteering, a Wizard-of-Oz setup in which one operator drives all three avatars so that behavior is held constant across conditions. Because all three avatars depict the same person (the experimenter), identity is held constant across conditions, which also makes it possible to add a real-world control condition in a later study.

5.1 Research questions and hypotheses

- RQ1: How does an avatar’s visual realism affect social presence, co-presence, interpersonal trust, and eeriness when the depicted identity and behavior are held constant?
- RQ2 (uncanny valley [37]): Is the relationship between realism and experience monotonic, or does the mid-realism avatar incur the greatest eeriness?



Figure 5: A stylized Meta avatar of the same person, built in Meta’s avatar tool from a phone capture and rendered as a cartoon figure for Horizon Worlds. It is the stylized end of the range we compare in the proposed study. Meta reaches a very large audience with avatars that make no attempt at photographic likeness, which is part of why the value of realism is still contested.

- **RQ3 (metric disconnect):** Do image-fidelity metrics (PSNR/SSIM/LPIPS) predict any of the experiential measures in RQ1 and RQ2?

As a manipulation check, we will also ask participants to rate the perceived visual realism and identity fidelity of each avatar.

We advance no directional hypothesis for RQ1. The literature in Section 2 supports both a benefit and a penalty of higher visual realism, and adjudicating between them is the purpose of the study. We do, however, state the following expectations:

- **H1:** Co-presence and interpersonal trust will differ across the three avatars, but not necessarily in a monotonic relationship with visual realism.
- **H2:** Eeriness will be non-monotonic across the three avatars, being greatest for the mid-realism crafted avatar [47, 34].
- **H3:** Image-fidelity metrics (PSNR, SSIM, LPIPS) will correlate at best weakly with the experiential measures.

5.2 Design

The study crosses three avatars with three environments in a fully within-subjects design. The avatars span the realism range: a *captured* avatar (3DGS/GaussianAvatars), a *crafted* avatar (Epic MetaHuman), and a *stylized* avatar (Meta/Horizon). Three different versions of the same real environment will be developed to matched the realism levels of the avatars: a captured (or indistinguishable-from-captured) environment, a photorealistically modeled environment, and a stylized environment. Crossing the two factors yields nine static scenes; every participant will view all nine, with the presentation order counterbalanced between participants. The avatar will be seated and its body organically occluded, for example by being located behind a table, to accommodate the limitations of the current capture pipeline. To address possible appearance confounds, we will build each avatar at the highest quality currently achievable with its production route, and will treat visual realism and identity fidelity as measured variables; the premise of the study is precisely that one cannot predict the experience of a level of realism that cannot yet be reached.

5.3 Visual realism and identity fidelity

We will assess visual realism and identity fidelity as separate constructs. Visual realism refers to how photorealistic the rendering

appears; identity fidelity refers to how faithfully the avatar resembles the specific person. The two are independent: a MetaHuman can look highly realistic yet not resemble a given person (high visual realism, low identity fidelity), while a stylized avatar may be easily recognized as a given person without appearing photographically realistic. Because all avatars depict the experimenter, whom participants will meet in person before the session, both constructs can be rated meaningfully, and we will analyze them separately.

5.4 Measures

After viewing each scene, participants will rate six measures: co-presence, social presence, interpersonal trust, eeriness, visual realism, and identity fidelity. We will draw items from the Networked Minds Measure of Social Presence [3] and the co-presence items of Bailenson et al. [2]; the revised uncanny-valley indices of Ho and MacDorman [20] for humanness and eeriness; a short likeness scale together with a photograph recognition test for identity fidelity [15]; and the Specific Interpersonal Trust Scale [23] for trust, a short version of which has VR precedent [38]. We will also assess spatial presence with the Igroup Presence Questionnaire [42]. For each rendering we will additionally compute PSNR, SSIM, and LPIPS against held-out real video of the person; these will serve as predictors for RQ3, not as outcome measures.

5.5 Participants and procedure

Participants: We plan to recruit 36 participants from our local university community. We determined this sample size by using an a priori power analysis in G*Power [12]: for the primary within-subjects comparison (difference between two dependent means, matched pairs; two-tailed paired t test) at a medium effect size ($d_z = 0.5$), $\alpha = .05$, and power = .80, the required sample is 34, which we round up to 36 to be conservative. We adopt the conventional benchmarks of a medium effect ($d_z = 0.5$) and power = .80, the value Cohen proposes for general use as both standard and feasible [8]. Participants will be screened for stereo vision and we will also record demographic data including prior VR experience.

Procedure: Participants will be briefed and give informed consent in person, so they will meet the experimenter in real life before encountering her again in VR. They will then put on the HMD and remain in it for the whole session. We will familiarize participants with the questions they will be asked ahead of time, so they will know what to attend to before beginning the first trial. Each of the nine scenes will be presented for about 1 minutes; participants will then answer the questions in situ, reading the question prompts which will be presented in a diegetic manner and speaking their responses aloud for the experimenter to record, so that no controllers or menus intrude on the experience. Scenes will fade to black between presentations. Participants will be seated and head-tracked, which preserves motion parallax, but will not walk around. We anticipate that the full session will last about 20-30 minutes, including onboarding.

5.6 Ethics

Participants will view a photorealistic likeness of a real person. They will need to give informed consent to participate and to have their responses recorded, and the depicted person has already consented to the use of her likeness. The study will be conducted under Institutional Review Board (IRB) approval, which will be obtained before recruitment begins.

6 DISCUSSION

No user-study results are reported here; this paper contributes a capture pipeline and a study design. The design compares one identity rendered at three levels of realism, each in the same environment, also rendered at three corresponding levels of realism, and it relates the experiential outcomes to image-fidelity metrics, addressing a

question the field has not settled: whether pursuing perfection-level photorealism in avatar depictions is a worthy aim for VR/XR applications. To our knowledge, this would be the first study to compare a captured radiance-field avatar, a crafted MetaHuman, and a stylized avatar of the same person in a within-subjects design, and to test whether image metrics predict how each is experienced.

Limitations and Future Work This study will use a single identity and static avatars. Adding motion and behavior is the natural next step, but it introduces a confound the present design avoids: imperfect facial tracking, lip-sync artifacts, or other shortcomings in the dynamics of the avatar behavior could dominate the experience regardless of the avatar appearance, and while we are on the cusp of being able to achieve truly photorealistic static virtual human models, the challenges are greater in achieving dynamic avatars that could be mistaken for real people. To make this more tractable, we plan to begin with animated avatars that are each controlled by the same captured motions obtained via live puppeteering. Future studies with dynamic avatars will additionally include a real-world control condition. Behavioral measures of trust are also left to future work.

7 ACKNOWLEDGMENTS

The authors acknowledge the Minnesota Supercomputing Institute (MSI) at the University of Minnesota for providing resources that contributed to the research results reported within this paper.

REFERENCES

- [1] S. Aseeri and V. Interrante. The influence of avatar representation on interpersonal communication in virtual social environments. *IEEE Transactions on Visualization and Computer Graphics*, 27:2608–2617, 2021. 2
- [2] J. N. Bailenson, N. Yee, D. Merget, and R. Schroeder. The effect of behavioral realism and form realism of real-time avatar faces on verbal disclosure, nonverbal disclosure, emotion recognition, and co-presence in dyadic interaction. *Presence: Teleoperators and Virtual Environments*, 15(4):359–372, 2006. doi: 10.1162/pres.15.4.359 2, 5
- [3] F. Biocca, C. Harms, and J. K. Burgoon. Toward a more robust theory and measure of social presence: Review and suggested criteria. *Presence: Teleoperators and Virtual Environments*, 12(5):456–480, 2003. doi: 10.1162/105474603322761270 2, 5
- [4] V. Blanz and T. Vetter. A morphable model for the synthesis of 3D faces. In *Proc. SIGGRAPH '99*, pp. 187–194. ACM Press/Addison-Wesley, 1999. doi: 10.1145/311535.311556 1, 2
- [5] J. Blascovich, J. Loomis, A. C. Beall, K. R. Swinith, C. L. Hoyt, and J. N. Bailenson. Immersive virtual environment technology as a methodological tool for social psychology. *Psychological Inquiry*, 13(2):103–124, 2002. doi: 10.1207/S15327965PLI1302.01 2
- [6] Blender Foundation. Blender. <https://www.blender.org>, 2026. Accessed 2026-07-11. 4
- [7] J. Chen, J. Hu, G. Wang, Z. Jiang, T. Zhou, Z. Chen, and C. Lv. TaoAvatar: Real-time lifelike full-body talking avatars for augmented reality via 3D gaussian splatting. In *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2
- [8] J. Cohen. A power primer. *Psychological Bulletin*, 112(1):155–159, 1992. doi: 10.1037/0033-2909.112.1.155 5
- [9] C. Dubosc, G. Gorisse, T. Lefrou, S. Richir, and O. Christmann. Effect of avatar stylization and facial expression intensity in virtual interactions. *Virtual Reality*, 29(4):163, 2025. doi: 10.1007/s10055-025-01238-6 3
- [10] Epic Games. MetaHuman documentation. <https://dev.epicgames.com/documentation/metahuman/metahuman-documentation>, 2026. Technical documentation. Accessed 2026-07-09. 2, 3, 4
- [11] Epic Games. Unreal engine. <https://www.unrealengine.com>, 2026. Accessed 2026-07-11. 4
- [12] F. Faul, E. Erdfelder, A.-G. Lang, and A. Buchner. G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences. *Behavior Research Methods*, 39(2):175–191, 2007. doi: 10.3758/BF03193146 5
- [13] A. D. Fraser, I. Branson, R. C. Hollett, C. P. Speelman, and S. L. Rogers. Do realistic avatars make virtual reality better? Examining human-like avatars for VR social interactions. *Computers in Human Behavior: Artificial Humans*, 2(2):100082, 2024. doi: 10.1016/j.chbah.2024.100082 3
- [14] G. Freeman and D. Maloney. Body, avatar, and me: The presentation and perception of self in social virtual reality. *Proc. ACM Hum.-Comput. Interact. (CSCW)*, 4(CSCW3), 2020. doi: 10.1145/3432938 2
- [15] M. C. Fysh, I. V. Trifonova, J. Allen, C. McCall, A. M. Burton, and M. Bindemann. Avatars with faces of real people: A construction method for scientific experiments in virtual reality. *Behavior Research Methods*, 54(3):1461–1475, 2022. doi: 10.3758/s13428-021-01676-5 5
- [16] G. Gafni, J. Thies, M. Zollhöfer, and M. Nießner. Dynamic neural radiance fields for monocular 4D facial avatar reconstruction. In *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2021. doi: 10.1109/CVPR46437.2021.00854 1, 2
- [17] M. Garau, M. Slater, V. Vinayagamoorthy, A. Brogni, A. Steed, and M. A. Sasse. The impact of avatar realism and eye gaze control on perceived quality of communication in a shared immersive virtual environment. In *Proc. SIGCHI Conf. on Human Factors in Computing Systems (CHI)*, pp. 529–536, 2003. doi: 10.1145/642611.642703 2
- [18] M. Gonzalez-Franco and T. C. Peck. Avatar embodiment: towards a standardized questionnaire. *Frontiers in Robotics and AI*, 5:74, 2018. doi: 10.3389/frobt.2018.00074 2
- [19] P. Heidicker, E. Langbehn, and F. Steinicke. Influence of avatar appearance on presence in social VR. In *Proc. IEEE Symp. on 3D User Interfaces (3DUI)*, pp. 233–234, 2017. doi: 10.1109/3DUI.2017.7893357 2
- [20] C.-C. Ho and K. F. MacDorman. Measuring the uncanny valley effect: Refinements to indices for perceived humanness, attractiveness, and eeriness. *International Journal of Social Robotics*, 9(1):129–139, 2017. doi: 10.1007/s12369-016-0380-9 5
- [21] L. Hu, S. Saito, L. Wei, K. Nagano, J. Seo, J. Fursund, I. Sadeghi, C. Sun, Y.-C. Chen, and H. Li. Avatar digitization from a single image for real-time rendering. *ACM Trans. Graph. (SIGGRAPH Asia)*, 36(6):195:1–195:14, 2017. doi: 10.1145/3130800.3130887 1, 2
- [22] A. E. Ichim, S. Bouaziz, and M. Pauly. Dynamic 3D avatar creation from hand-held video input. *ACM Trans. Graph. (SIGGRAPH)*, 34(4):45:1–45:14, 2015. doi: 10.1145/2766974 1, 2
- [23] C. Johnson-George and W. C. Swap. Measurement of specific interpersonal trust: Construction and validation of a scale to assess trust in a specific other. *Journal of Personality and Social Psychology*, 43(6):1306–1317, 1982. doi: 10.1037/0022-3514.43.6.1306 5
- [24] J.-N. Kaiser, S. Kimmel, E. Licht, E. Landwehr, F. Hemmert, and W. Heuten. Get real with me: Effects of avatar realism on social presence and comfort in augmented reality remote collaboration and self-disclosure. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*, pp. 1–18. Association for Computing Machinery, 2025. doi: 10.1145/3706598.3713541 3
- [25] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph. (SIGGRAPH)*, 42(4):139:1–139:14, 2023. doi: 10.1145/3592433 1, 2
- [26] K. Kiltén, R. Groten, and M. Slater. The sense of embodiment in virtual reality. *Presence: Teleoperators and Virtual Environments*, 21(4):373–387, 2012. doi: 10.1162/PRES.a.00124 2
- [27] M. E. Latoschik, D. Roth, D. Gall, J. Achenbach, T. Waltemate, and M. Botsch. The effect of avatar realism in immersive social virtual realities. In *Proc. ACM Symp. on Virtual Reality Software and Technology (VRST)*, 2017. doi: 10.1145/3139131.3139156 2
- [28] T. Li, T. Bolkart, M. J. Black, H. Li, and J. Romero. Learning a model of facial shape and expression from 4D scans. *ACM Trans. Graph. (SIGGRAPH Asia)*, 36(6):194:1–194:17, 2017. doi: 10.1145/3130800.3130813 1, 2, 3
- [29] S. Lin, L. Yang, I. Saleemi, and S. Sengupta. Robust high-resolution video matting with temporal guidance. In *Proc. IEEE/CVF Winter Conf. on Applications of Computer Vision (WACV)*, pp. 238–247,

2022. doi: 10.1109/WACV51458.2022.00319 3
- [30] S. Lombardi, J. Saragih, T. Simon, and Y. Sheikh. Deep appearance models for face rendering. *ACM Trans. Graph. (SIGGRAPH)*, 37(4):68:1–68:13, 2018. Foundation of Meta’s Codec Avatars. doi: 10.1145/3197517.3201401 1, 2
- [31] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. SMPL: A skinned multi-person linear model. *ACM Trans. Graph. (SIGGRAPH Asia)*, 34(6):248:1–248:16, 2015. doi: 10.1145/2816795.2818013 1, 2
- [32] S. Ma, T. Simon, J. Saragih, D. Wang, Y. Li, F. De la Torre, and Y. Sheikh. Pixel codec avatars. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 64–73, June 2021. 2
- [33] K. Mania, B. D. Adelstein, S. R. Ellis, and M. I. Hill. Perceptual sensitivity to head tracking latency in virtual environments with varying degrees of scene complexity. In *Proc. 1st Symposium on Applied Perception in Graphics and Visualization (APGV)*, pp. 39–47. ACM, 2004. doi: 10.1145/1012551.1012559 2
- [34] R. McDonnell, M. Breidt, and H. H. Bülthoff. Render me real? Investigating the effect of render style on the perception of animated virtual humans. *ACM Trans. Graph. (SIGGRAPH)*, 31(4):91:1–91:11, 2012. doi: 10.1145/2185520.2185587 2, 3, 5
- [35] T. Menzel, E. Wolf, S. Wenninger, N. Spinczyk, L. Holderrieth, C. Wienrich, U. Schwanecke, M. E. Latoschik, and M. Botsch. Avatars for the masses: Smartphone-based reconstruction of humans for virtual reality. *Frontiers in Virtual Reality*, 6:1583474, 2025. doi: 10.3389/frvir.2025.1583474 3
- [36] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. NeRF: Representing scenes as neural radiance fields for view synthesis. In *Proc. European Conf. on Computer Vision (ECCV)*, 2020. doi: 10.1007/978-3-030-58452-8_24 1, 2
- [37] M. Mori, K. F. MacDorman, and N. Kageki. The uncanny valley [from the field]. *IEEE Robotics & Automation Magazine*, 19(2):98–100, 2012. Original work published 1970. doi: 10.1109/MRA.2012.2192811 4
- [38] Y. Pan and A. Steed. The impact of self-avatars on trust and collaboration in shared virtual environments. *PLOS ONE*, 12(12):e0189078, 2017. doi: 10.1371/journal.pone.0189078 5
- [39] Polycam Inc. Polycam - 3d scanner & lidar. Available at <https://poly.cam/>, 2026. Mobile application software. Accessed 2026-07-11. 4
- [40] S. Qian, T. Kirschstein, L. Schoneveld, D. Davoli, S. Giebenhain, and M. Nießner. GaussianAvatars: Photorealistic head avatars with rigged 3D gaussians. In *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, pp. 20299–20309, 2024. doi: 10.1109/CVPR52733.2024.01919 1, 2, 3
- [41] K. Raaen and I. Kjellmo. Measuring latency in virtual reality systems. In *Entertainment Computing – ICEC 2015*, vol. 9353 of *Lecture Notes in Computer Science*, pp. 457–462. Springer, 2015. doi: 10.1007/978-3-319-24589-8_40 2
- [42] T. Schubert, F. Friedmann, and H. Regenbrecht. The experience of presence: Factor analytic insights. *Presence: Teleoperators and Virtual Environments*, 10(3):266–281, 2001. doi: 10.1162/105474601300343603 2, 5
- [43] M. Seymour, L. Yuan, A. R. Dennis, and K. Riemer. Have we crossed the uncanny valley? Understanding affinity, trustworthiness, and preference for realistic digital humans in immersive environments. *Journal of the Association for Information Systems*, 22(3):591–617, 2021. doi: 10.17705/jais.00674 3
- [44] M. Slater. Place illusion and plausibility can lead to realistic behaviour in immersive virtual environments. *Philosophical Transactions of the Royal Society B*, 364(1535):3549–3557, 2009. doi: 10.1098/rstb.2009.0138 2
- [45] M. Slater, D. Banakou, A. Beacco, J. Gallego, F. Maciá-Varela, and R. Oliva. A separate reality: An update on place illusion and plausibility in virtual reality. *Frontiers in Virtual Reality*, 3:914392, 2022. doi: 10.3389/frvir.2022.914392 2
- [46] A. Steed, Y. Pan, F. Zisch, and W. Steptoe. The impact of a self-avatar on cognitive load in immersive virtual reality. In *Proc. IEEE Virtual Reality (VR)*, pp. 67–76, 2016. doi: 10.1109/VR.2016.7504689 2
- [47] Z. Tao, Y. Liu, J. Qiu, and S. Li. Impact of virtual avatar appearance realism on perceptual interaction experience: A network meta-analysis. *Frontiers in Psychology*, 16:1624975, 2025. doi: 10.3389/fpsyg.2025.1624975 3, 5
- [48] Unity Technologies. Unity. <https://unity.com>, 2026. Accessed 2026-07-11. 3
- [49] N. Yee and J. Bailenson. The Proteus effect: The effect of transformed self-representation on behavior. *Human Communication Research*, 33(3):271–290, 2007. doi: 10.1111/j.1468-2958.2007.00299.x 2
- [50] R. Zhang, A. Watkins, and N. Sarkar. GSAC: Leveraging gaussian splatting for photorealistic avatar creation with unity integration. *arXiv preprint arXiv:2504.12999*, 2025. Also in IEEE Xplore. 2